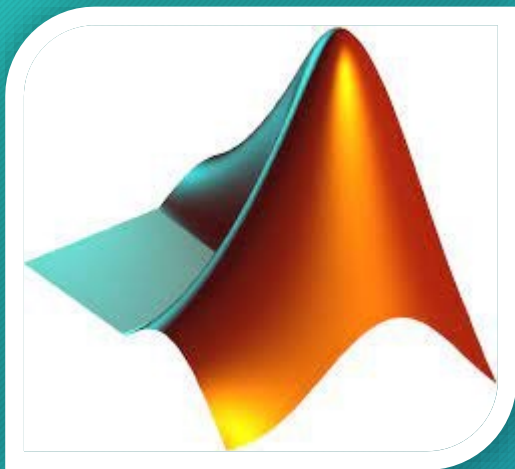




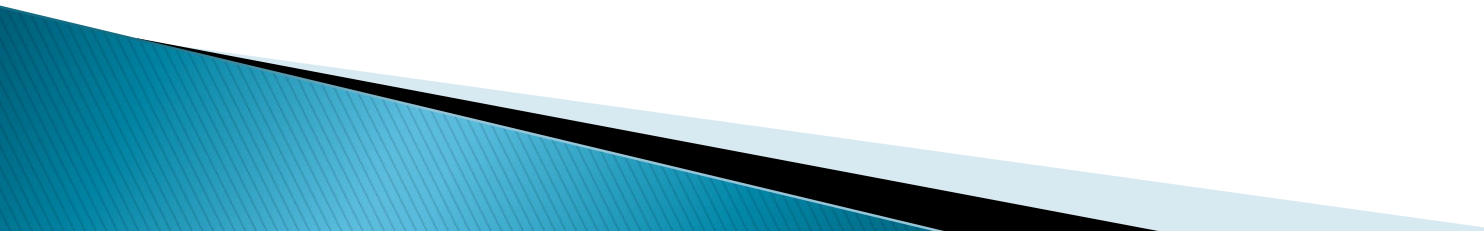
Communication Design

LAB.2

Signal Processing Overview

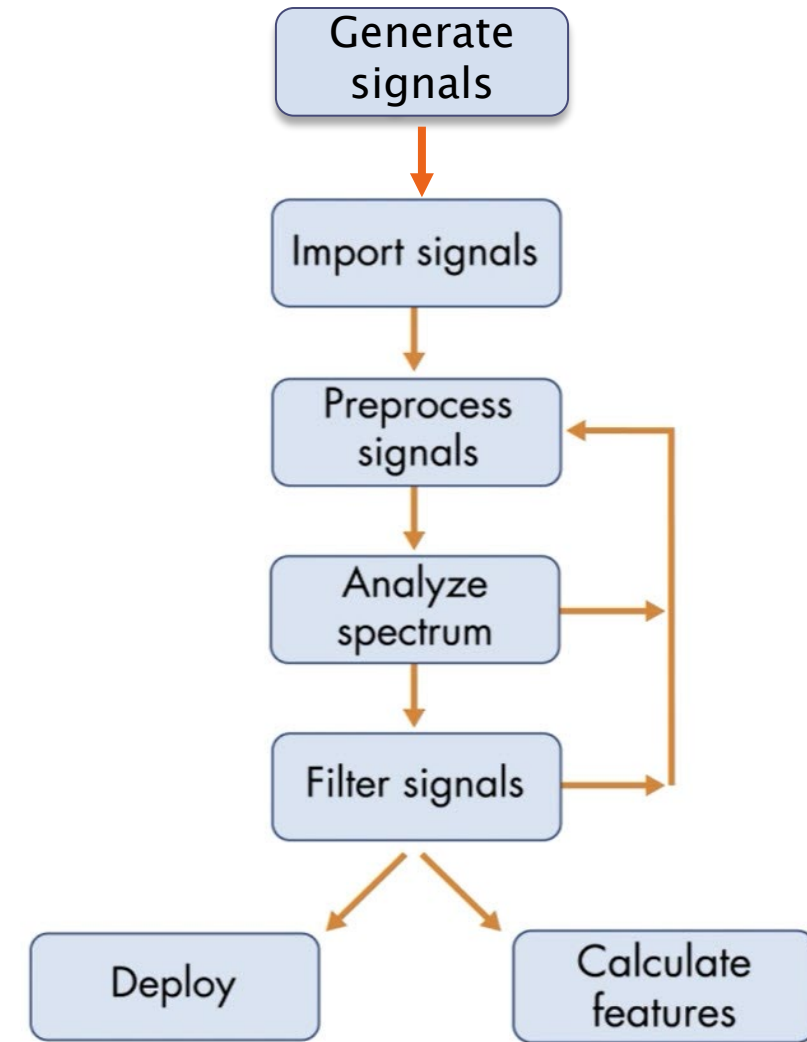
Ass.Lec.Ammar Alabbassi

Outline:

- ▶ Signal Processing Overview
 - ▶ Generate Signals
 - ▶ Import Signals
 - ▶ View Power Spectrum
 - ▶ Compare Signals
- 

Signal Processing Overview

1. Generate signals.
2. Import the signals into MATLAB.
3. Preprocess the signals in the time domain to prepare them for analysis
4. Look at the power spectrum in the frequency domain
5. Filter the signals to keep only relevant frequency content
6. Revisit earlier steps in the process to refine your algorithm
7. Extract information from the signals or deploy your algorithm to hardware



Generate Signals

Task 1

To generate a signal, you first choose the sample times for each point of the signal. You can choose sample times by specifying the following:

- Sample rate f_s in Hertz (Hz)
- Start time in seconds
- End time in seconds

You can create time steps from a to b with the colon operator:

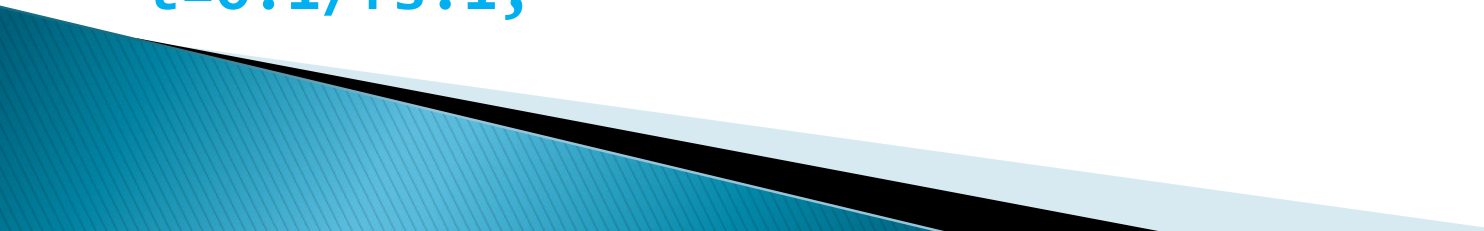
$a:1/f_s:b$

TASK

Specify a sample rate of 100 Hz in a variable named f_s . Then create a signal named t that extends from 0 seconds to 1 second, with time steps that are $1/f_s$.

```
 $f_s=100;$ 
```

```
 $t=0:1/f_s:1;$ 
```



Task 2

A straightforward way to make a sine wave is using the **sin** function:

```
s = sin(2*pi*f*t)
```

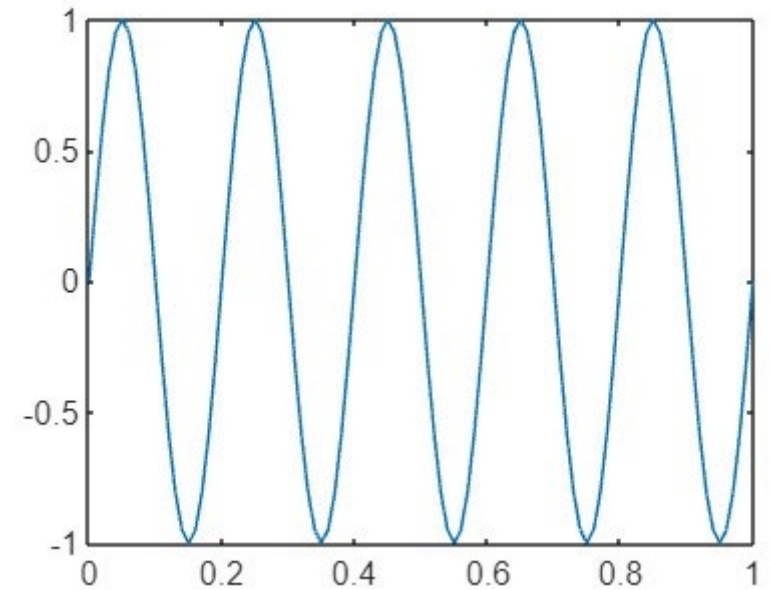
where f is the signal's frequency, and t are the time steps.

TASK

Create a 5 Hz sine signal named `sig` at the sample times in `t`.

Then create a plot of `sig` versus `t`.

```
sig = sin(2*pi*5*t);  
plot(t,sig)
```



Task 3

It is common to add noise to generated signals. Noise makes the signals more realistic and helps you to test if your signal processing techniques are robust to noisy data.

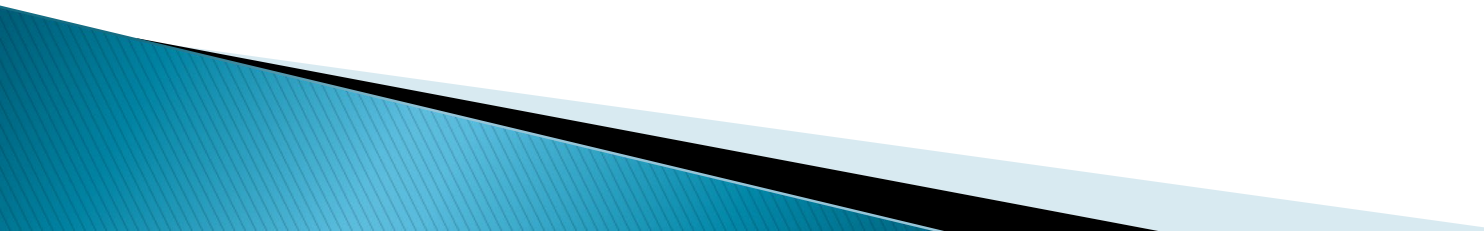
To add noise, you can create a vector of random numbers that is the same size as your signal **S**:

```
randn(size(s))
```

#TASK

Create a vector of normally distributed random numbers named **noise** that is the same size as **sig**. Multiply the vector by 0.1 to scale the vector.

```
noise=0.1*randn(size(sig));
```

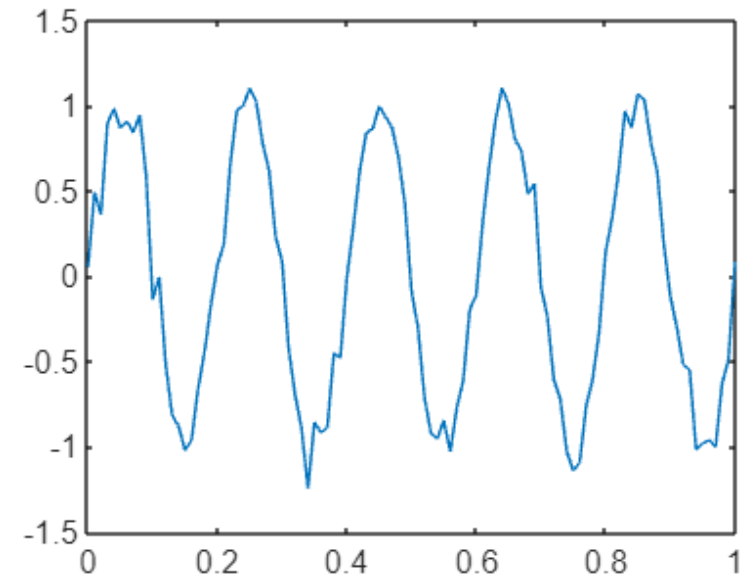


Task 4

#Task:

Add the signal vector with the noise vector. Name the output **sigNoisy**. Then create a plot of **sigNoisy** versus **t**.

```
sigNoisy = sig + noise;  
plot(t,sigNoisy)
```



Task 5

Signal Processing Toolbox contains functions for common waveforms, like the square and **sawtooth** functions. The inputs are similar to the **sin** function:

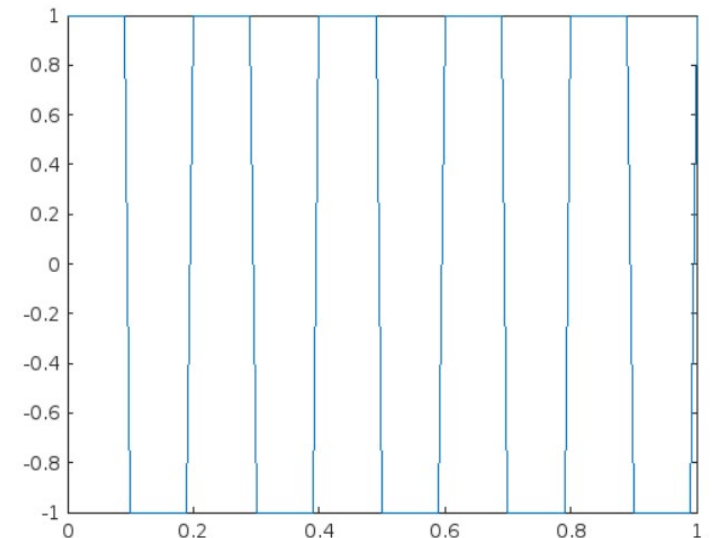
```
s = square(2*pi*f*t)
```

f is the frequency and t is a vector of time steps.

#TASK

Create a square signal named sq at the sample times in t. Use a frequency of 5. Then create a plot of sq versus t.

```
sq = square(2*pi*5*t);  
plot(t,sq)
```



Import Signals

Task 1

In this task, you will process the seismic signals in the Signal Analyzer app. The seismic stations record the vertical displacement caused by vibrations in the Earth's crust. First, you need to import the signals into the Workspace. The readmatrix function imports a numeric matrix from a file.

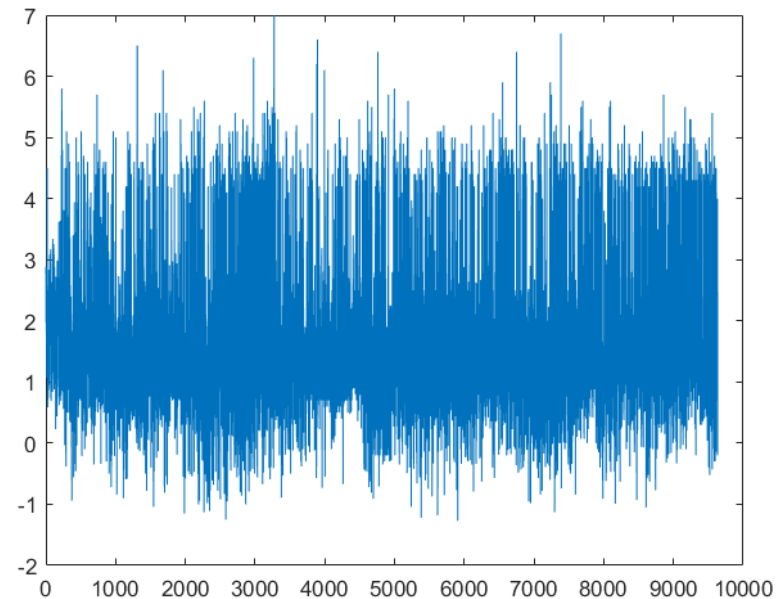
```
M = readmatrix("mynum.txt")
```

In this activity, you import the signal recorded at the HARP seismic station. The signal is stored in the file "harp.csv".

TASK

Import the seismic signal stored in "harp.csv" to a vector named harp. Then plot the harp signal.

```
harp = readmatrix("harp.csv")  
plot(harp)
```



View Power Spectrum

HOME PLOTS APPS EDITOR PUBLISH VIEW

Design App Get More Apps Install App Package App

Curve Fitter Optimization PID Tuner System Identification Wireless Waveform Ge... Signal Analyzer Instrument Control SimBiology Model Builder SimBiology Model Analyzer MATLAB Coder Application Compiler Class Diagram Viewer

FILE APPS

E: \ LECTURES 2023-2024 \ 2nd semester \ Communication Design Lab

Current Folder

- Name
- Assignments
- Fast Fourier
- Filters
- all_month.csv
- harp.csv
- harp.tif
- Lab1- Introduction.pptx
- New Microsoft Word Document.docx

Details

Editor - untitled3

untitled3

Workspace

Name

- harp
- t

Command History

```
t=1:1:126;  
plot(t,x)  
x=abs(randn(126,  
2x harp = readmatri  
clear  
clc
```

Columns 9,609 through 9,616

9609	9610	9611	9612	9613	9614	9615	9616
------	------	------	------	------	------	------	------

Select a file to view details

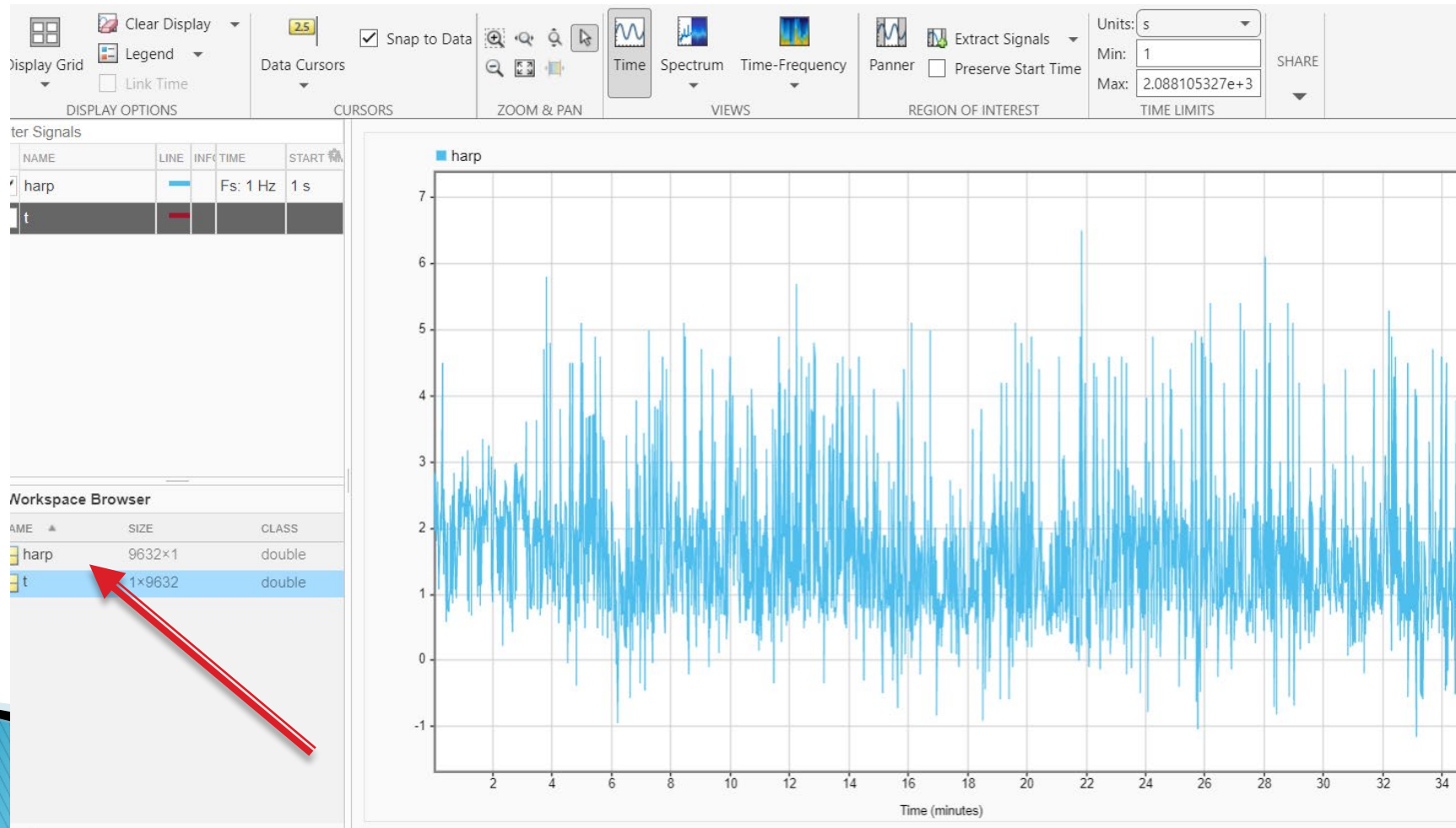
Task 1

The **Workspace Browser** within Signal Analyzer contains any signals stored as numeric arrays or variables with inherent time information like timetables.

You can visualize signals by dragging them from the **Workspace Browser** to the axes.

#TASK

Drag the **harp** signal from the Workspace Browser to the axes to display it.



Task 2

To set the time information, select **Time Values** in the **Analyzer** tab. You can specify the sample rate (fs), the time between each sample ($1/fs$), or a vector of time values.

#TASK

Set sample rate to 50 Hz. (In the Time Specification drop down, select Sample Rate and Start Time).

The screenshot shows the Signal Analyzer software interface. The main window has a menu bar with ANALYZER, DISPLAY, TIME, and MEASUREMENTS. Below the menu bar is a toolbar with icons for New, Open, Save, Display Grid, Signal Table, Workspace Browser, Measurements, Time Values, Export, Duplicate, Delete, Preprocess, Undo Preprocessing, Generate Function, and Preferences. The TIME tab is selected, and the Time Values dialog box is open. The dialog box has a title bar with a question mark and a close button. The main text in the dialog box is "Edit sample rate, sample time, or time values vector of selected signals". Below this is a "Time Specification:" label followed by a dropdown menu set to "Sample Rate and Start Time". There are two input fields: "Sample Rate:" with the value "50" and a unit dropdown set to "Hz", and "Start Time:" with the value "0" and a unit dropdown set to "s". At the bottom of the dialog box are "OK" and "Cancel" buttons. Red arrows point to the "Time Values" button in the toolbar, the "Sample Rate and Start Time" dropdown, the "50" input field, and the "0" input field.

Signal Analyzer - untitled*

ANALYZER DISPLAY TIME MEASUREMENTS

FILE LAYOUT SIGNAL TABLE PREPROCESSING OPTIONS

Filter Signals

NAME	LINE	INFO	TIME	START
harp			Fs: 50 ...	0 s

Workspace Browser

NAME	SIZE	CLASS
harp	9632×1	double
t	1×9632	double

Time Values

Edit sample rate, sample time, or time values vector of selected signals

Time Specification: Sample Rate and Start Time

Sample Rate: 50 Hz

Start Time: 0 s

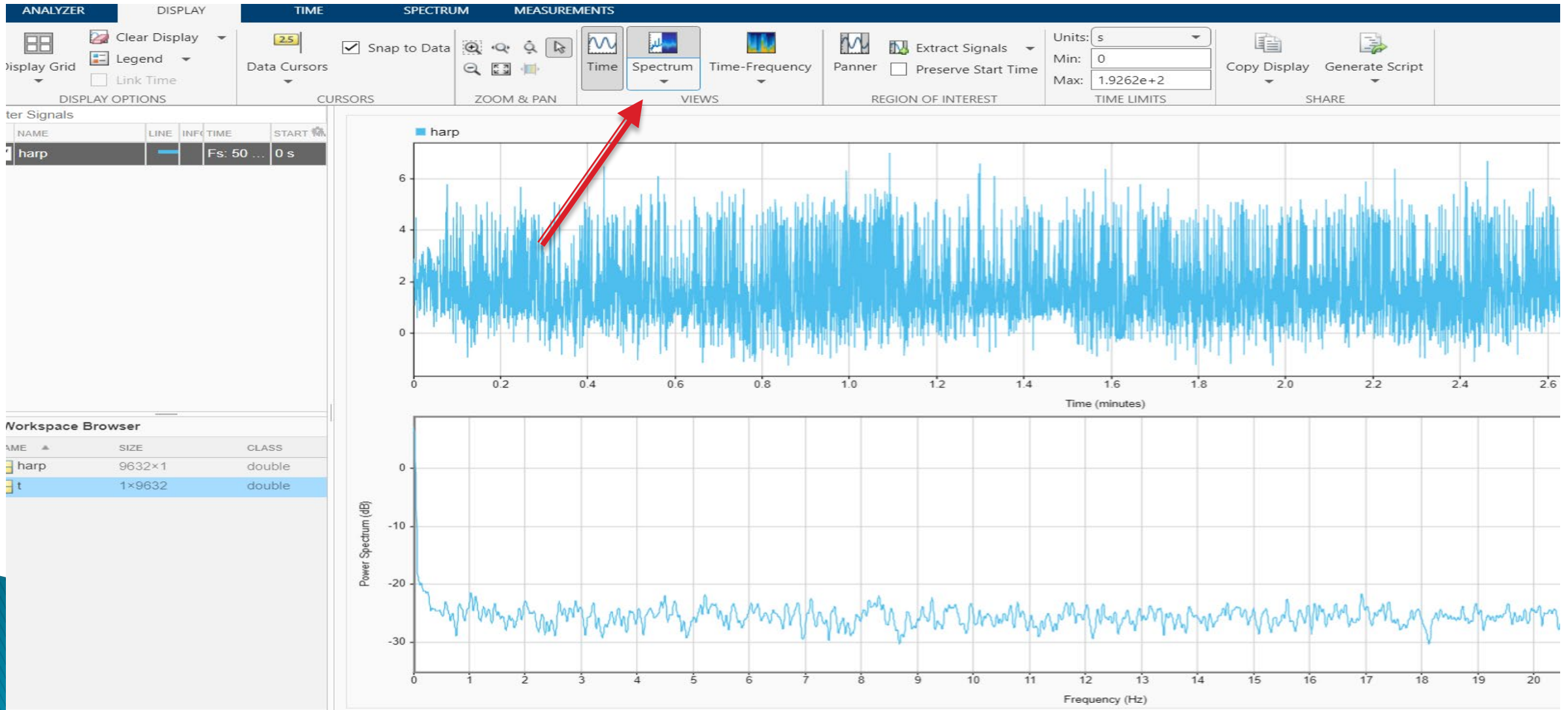
OK Cancel

Task 3

To view the frequency content of a signal, you can display its power spectrum by clicking **Spectrum** on the **Display** tab.

#TASK

Display the power spectrum of the harp signal.



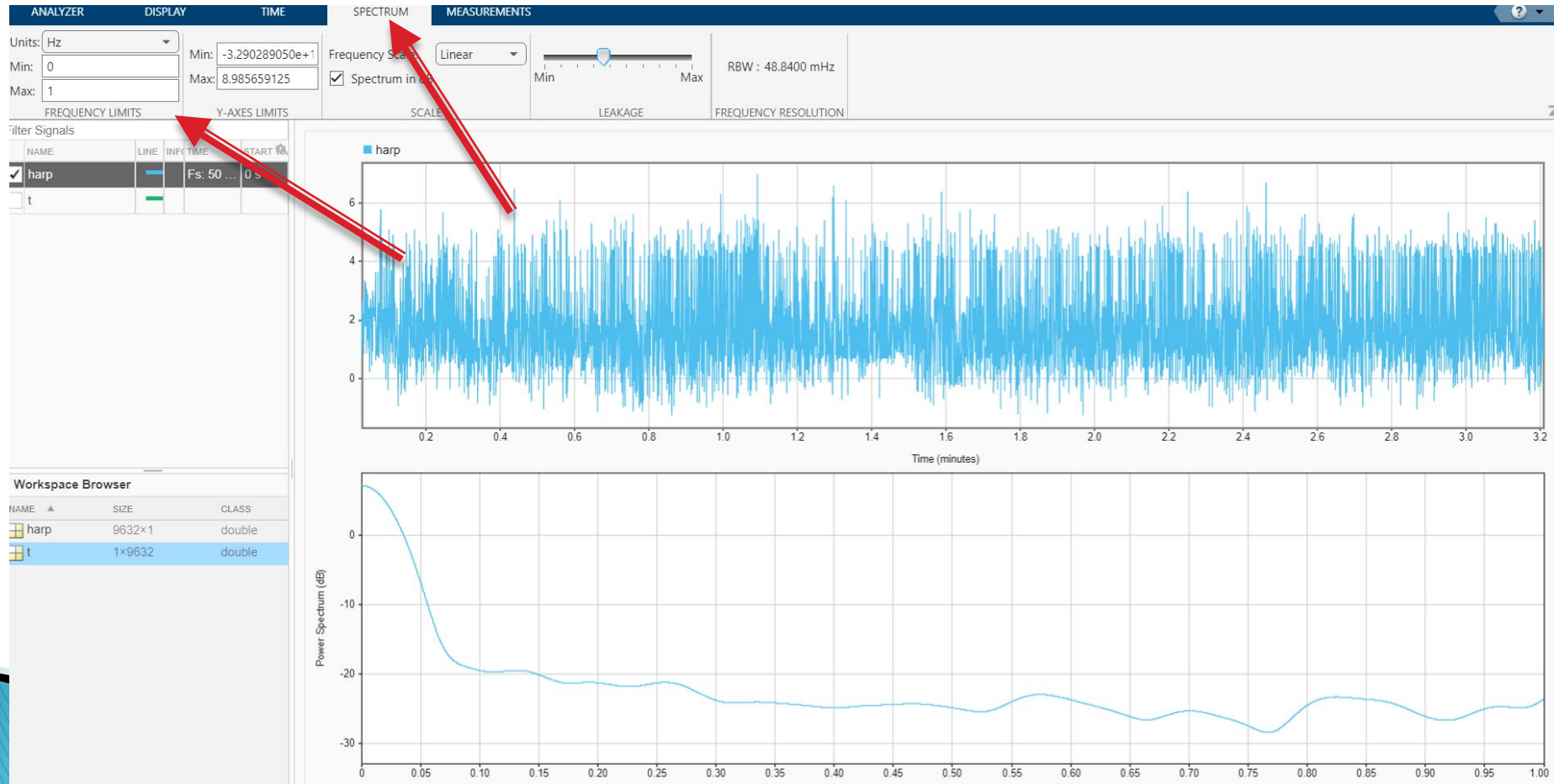
Task 4

While you display the power spectrum, a **Spectrum** tab contains options to customize the visualization.

The frequency range of these seismic surface waves is small, so to focus on the meaningful seismic activity, you can set the frequency limits of the power spectrum.

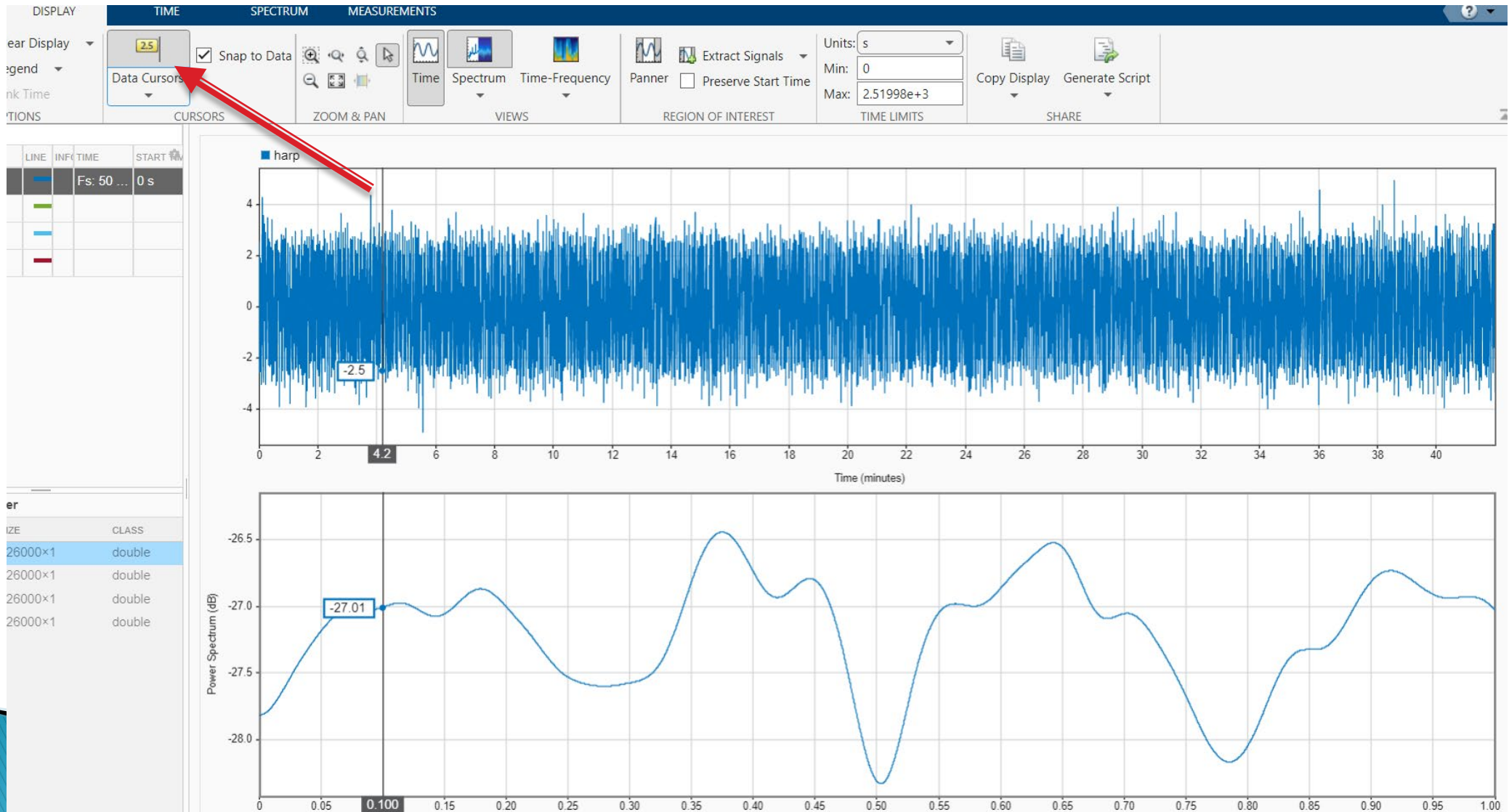
#TASK

Set the frequency limits from 0 Hz to 1 Hz.



Task 4

Try adding a data cursor to look at the frequency location of the peak. You can add a data cursor in the Cursors section of the Display tab.



Compare Signals

Task 1

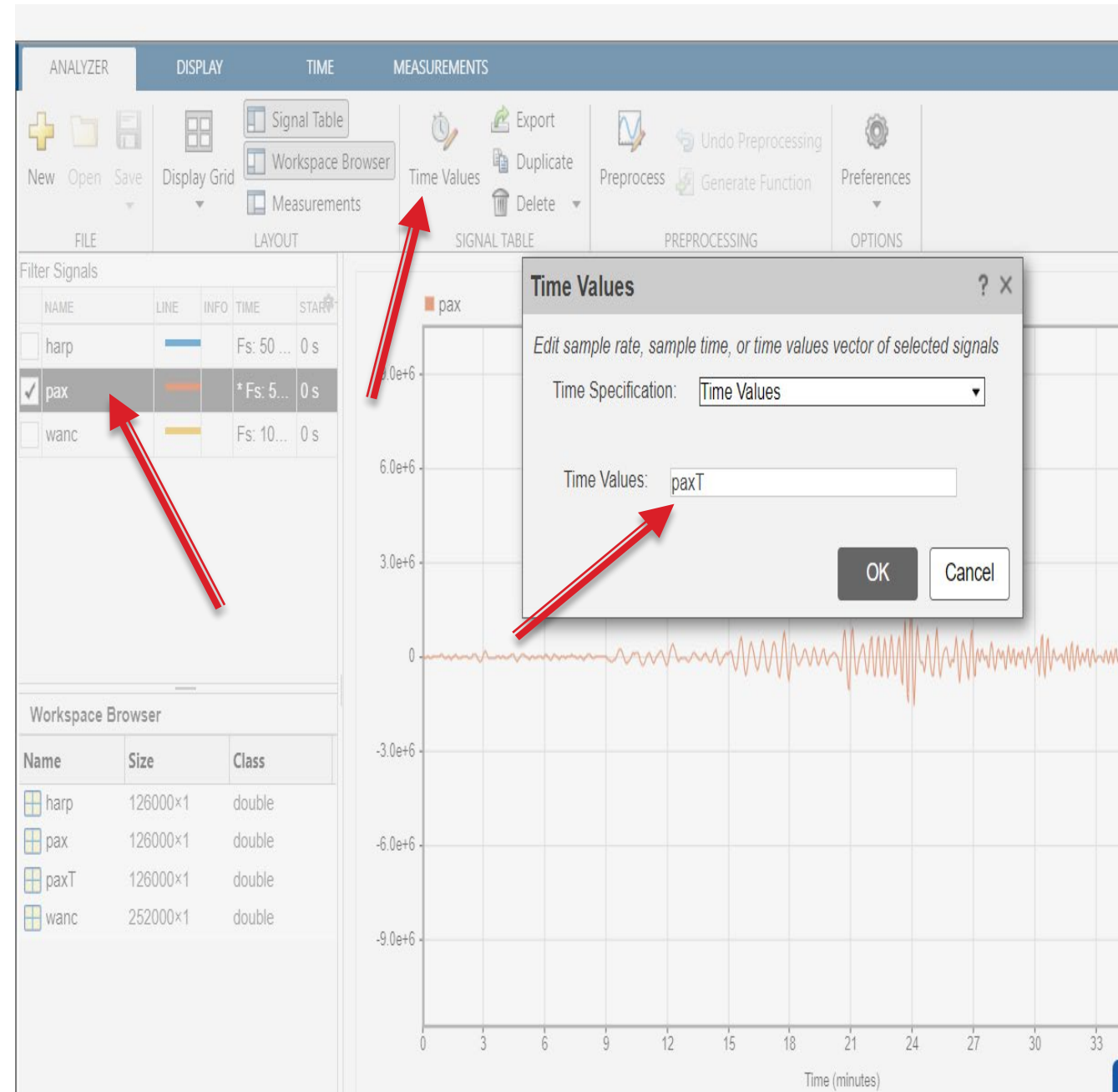
1-The three seismic stations **HARP**, **PAX**, and **WANC** collected the seismic signals using different hardware. As you've seen, the data from the **HARP** station was sampled at **50 Hz**.

2-The data from the **PAX** station was also sampled at **50 Hz**, but the hardware was **imperfect**, so the spacing between each sample wasn't exactly **0.02 seconds**. Since the time information of the signal isn't evenly spaced, the time information at every sample needs to be specified.

3- You can specify a vector instead of a sample rate using the Time Values time specification.

#TASK

Select the **pax** signal in the Signal table. Use the Time Values button to add time information using the vector **paxT**. Then display the signal.



Task 2

The signal from the **WANC** station is uniformly spaced, but it was sampled at a faster sample rate of **100 Hz**.

#TASK

Set the sample rate of the **wanc** signal to **100 Hz** and display the signal.

The screenshot displays a software interface with a top toolbar containing icons for New, Open, Save, Display Grid, Signal Table, Workspace Browser, Measurements, Export, Duplicate, Delete, Preprocess, Undo Preprocessing, Generate Function, and Preferences. Below the toolbar is a 'Filter Signals' table with columns NAME, LINE, INFO, TIME, and START. The 'wanc' signal is selected. Below this is a 'Workspace Browser' table with columns Name, Size, and Class.

NAME	LINE	INFO	TIME	START
harp		Fs: 50 ...	0 s	
pax		* Fs: 5...	0 s	
✓ wanc				

Name	Size	Class
harp	126000×1	double
pax	126000×1	double
paxT	126000×1	double
wanc	252000×1	double

The 'Time Values' dialog box is open, showing the 'Sample Rate and Start Time' specification. The 'Sample Rate' is set to 100 Hz, and the 'Start Time' is set to 0 s. The background shows a signal plot for the 'wanc' signal.

Task 3

The different signal amplitudes make it difficult to compare the signals in the **time domain**, but you can still compare them in the **frequency domain**.

#TASK

Display **all three signals**, then display the **power spectra**. Set the **frequency limits from 0 to 1 Hz**.

